

MARILYN, Antarctica

WMO: 898690

Lat: 79.904S

Lon: 165.774E

Elev: 62

StdP: 100.58

Time Zone: 12.00 (NZL)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-51.4	-48.4	N/A	N/A	N/A	N/A	N/A	N/A	19.2	-28.9	17.3	-28.0	2.4	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.7	1.0	N/A	-0.7	N/A	-1.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.1	12.0	DB	-55.0	6.4	2.8	2.4	-57.0	8.1	-58.7	9.5	-60.2	10.8	-62.3	12.6	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-22.7	-6.4	-15.8	-25.1	-28.4	-30.1	-30.0	-31.9	-32.2	-29.4	-23.7	-13.9	-5.6
	DBStd	11.16	3.32	5.02	5.57	6.41	7.71	7.03	7.25	7.48	7.08	4.96	5.07	2.83
	HDD10.0	11946	507	722	1088	1153	1242	1199	1298	1309	1181	1045	716	484
	HDD18.3	14987	766	956	1346	1403	1500	1449	1557	1568	1431	1303	966	743
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.0	2.8	3.6	5.3	6.1	5.9	5.8	6.6	5.6	6.2	5.4	3.6	2.6
Precipitation	PrecAvg	175	18	17	15	13	14	15	12	15	14	13	14	15
	PrecMax	282	43	37	40	38	32	41	38	53	34	36	36	30
	PrecMin	58	3	5	3	5	2	3	1	1	1	1	3	1
	PrecStd	48	11	9	7	7	8	10	8	12	9	9	8	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	-2.4	-9.3	-8.8	-10.4	-10.8	-12.0	-12.2	-9.4	-9.5	-2.6	5.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	1.1	-5.6	-12.6	-14.1	-13.8	-14.4	-15.5	-15.8	-13.6	-11.8	-4.1	1.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	-0.6	-7.3	-14.7	-16.9	-17.0	-17.4	-18.9	-18.3	-17.3	-13.6	-5.6	-0.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	-1.8	-8.6	-16.7	-19.2	-19.8	-20.1	-22.0	-21.4	-20.2	-15.9	-7.0	-1.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	7.9	7.8	8.6	8.7	8.2	9.3	8.3	8.6	8.2	7.7
		MCDBR	9.3	8.8	8.2	10.3	11.1	10.0	9.4	10.5	9.1	10.5	8.6	9.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.230	0.208	0.162	N/A	N/A	N/A	N/A	N/A	0.153	0.190	0.211	0.228
	taud		2.183	2.189	2.094	N/A	N/A	N/A	N/A	N/A	2.052	2.179	2.199	2.192
	Ebn at Noon		952	882	716	0	0	0	0	0	713	908	977	982
	Edh at Noon		99	75	41	0	0	0	0	0	41	76	97	106
All-Sky Solar Radiation	RadAvg		8.19	4.80	1.40	0.00	0.00	0.00	0.00	0.00	0.66	3.49	7.67	9.55
	RadStd		0.25	0.21	0.07	0.00	0.00	0.00	0.00	0.00	0.04	0.12	0.24	0.24

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page